Exam 4 Worksheet

CARDIAC

Cardiac Terms

- Define each of the following terms:
  - Afterload:
  - Preload:
  - Cardiac output:
  - Systole:
  - Diastole:
- Explain blood pressure using the terms systole and diastole.

Hypertension

- What occurs when BP is too low?
- What occurs when BP is too high?
- List 5 risk factors for developing HTN:
  -
  -
  -
  -
  -
- What are target-organs? List 3 examples of target-organs and the associated complications for each:
  -
  -
  -
- What is a hypertension crisis? What S/S would you expect to see?

Atherosclerosis

- Atherosclerosis is commonly found in patients with _____.
- What is the difference between LDL and HDL?
- Name 5 risk factors for developing atherosclerosis:
  -
  -
  -
  -
  -
- What are the five steps involved in the process of atherosclerosis?
  -
  -
  -
  -
  -
- What is the difference between a stable and unstable plaque?
- Explain how atherosclerosis can cause total vessel obstruction.

Peripheral Vascular Disease

- Complete the chart below:

	Peripheral Arterial Disease	Venous Insufficiency
Risk Factors		
Leg Color		
Ulcer/Wound Characteristics		
Distal Pulse Strength		
Leg Temperature		
Paresthesias ?		
Complications		

- What is intermittent claudication? Does the pain get better with rest?

Aneurysms

- Name three types of true aneurysms:
  - o
  - o
  - o
- What is an aneurysm dissection?
  - o An aneurysm dissection is most likely to occur in the _____.
- List three risk factors for aneurysm development:
  - o
  - o
  - o
- What signs and symptoms would you expect to see in a patient with a dissecting aortic aneurysm? Explain the complications you expect to see.

Myocardial Infarction

- An MI is detected by _____ and _____.
- Describe the signs and symptoms you expect to see in a patient with angina or MI.
- What is the brief pathophysiology involved in angina?
  - o When does angina typically occur?

    - List two differences between stable and unstable angina:
        -
        -
- Unscramble the underlined words:
    - Angina involves C E I S H M A I
    - MI involves C E S I R S N O
- List two differences between a STEMI and NSTEMI:
    -
    -
- What complications are seen in an MI that are not seen in angina?

Pericardial Effusion

- What is a pericardial effusion?
- Why may a patient with a large pericardial effusion have less signs and symptoms than a patient with a small pericardial effusion?
- What signs and symptoms are expected in a patient with a pericardial effusion?
- What is the main complication of a pericardial effusion?

Infective Endocarditis

- What is endocarditis?
- Name three patient populations that are at risk for infective endocarditis:
    -
    -
    -
- Name two signs and symptoms that are unique to infective endocarditis:
- What is a complication of infective endocarditis?

CHF

- Regardless if it is a diastolic or systolic issue, CHF always results in decreased _____ _____.
- List three disorders that we have discussed that can cause CHF. Explain the patho as to why they lead to CHF.
    -
    -
    -
- What specific lab value is elevated in CHF patients? (Hint: it is a hormone that measures hypervolemia, so it is also high in CKD patients)
- Describe the manifestations of right and left heart failure.
    - Right -
    - Left -
    - Why are they different?
- What is ejection fraction? What is the normal value?

- Circle the correct underlined words: EF is high/low in systolic/diastolic heart failure.

GI/HEPATOBILIARY

Nutrition

- What is BMI?
  - Match the terms below with it's associated BMI:

Underweight	24.6
Normal BMI	14.8
Overweight	32
Obese	27.5

- What is BMR?
  - Do males or females have a higher BMR?
- Explain the cause of weight loss and weight gain using the terms "food intake" and "energy expenditure."

GERD

- What is reflux/regurgitation?
- What differentiates occasional reflux from GERD?
- What are the two pathophysiological causes of GERD? List two associated risk factors for each cause.
  -
  -
- Describe the pain that patients with GERD present with.
- What is a complication of GERD?

Peptic Ulcer Disease (PUD)

- Which type of gastritis results in PUD?
- What are the two pathophysiological causes of PUD? List two associated risk factors for each cause.
  -
  -
- Why is it difficult for peptic ulcers to heal?
- Where are peptic ulcers found?
- Describe the pain that patients with PUD present with.
- List two complications of PUD and explain why they can occur:
  -
  -

Vomiting, Constipation, Diarrhea

- List three potential complications of vomiting and explain why they can occur:
  -

    - o
    - o
- List five potential risk factors for constipation:
    - o
    - o
    - o
    - o
    - o
- List three potential complications of diarrhea and explain why they can occur:
    - o
    - o
    - o
- What is the difference between melena and hematochezia? What disorders are usually associated with each?

Inflammatory Bowel Disease (IBD)

- What is the cause of IBD?
- What do exacerbations of IBD consist of?
- Which form of IBD results in a cobblestone appearance?
- Patients with which form of IBD are more at risk for malnutrition? Fluid and electrolyte imbalances? Why?
    - Malnutrition -
    - Fluid/electrolyte imbalance -
    - Why -
- Which form of IBD is associated with more complications? List two of these complications.
- Which form of IBD is associated with an increased risk of colon cancer?
- Which form of IBD is found anywhere along the GI tract in “skip lesions”?

Appendicitis

- Explain the pathophysiology of appendicitis. Include the manifestations and complications.

Small Bowel Obstruction (SBO)

- What are the two types of SBO? Explain them and list two potential causes of each:
    - o
    - o
- List five manifestations of a patient with an SBO and explain the pathophysiology as to why those occur:
    - o
    - o

    - o
    - o
    - o
- List three potential complications of SBO:
    - o
    - o
    - o

Hepatitis

- Which types of hepatitis are transmitted via blood?
- Which types of hepatitis are chronic (and thus at increased risk for developing cirrhosis and liver cancer)?
- List three patient populations that are at increased risk of acquiring hepatitis C:
    - o
    - o
    - o
- For which types of hepatitis do vaccines exist?
- What are the three stages of acute hepatitis and what occurs in each stage?
    - o
    - o
    - o

Cirrhosis

- List two potential causes of cirrhosis:
    - o
    - o
- What is portal hypertension?
    - Explain the pathophysiology of how ascites develops.
    - What are two other complications of portal hypertension?
- Explain why cirrhotic patients are jaundiced (icteric).
    - What other manifestations would you expect to see in an icteric patient?
- What is the complication involved with increased ammonia levels? How do these patients present?
- Circle the expected lab values for a cirrhotic patient and explain why:
    - PT (high/normal/low) -
    - PTT (high/normal/low) -
    - INR (high/normal/low) -
    - Platelets (high/normal/low) -
    - ALT (high/normal/low) -
    - AST (high/normal/low) -
    - Alkaline phosphatase (high/normal/low) -
    - Ammonia (high/normal/low) -
    - Bilirubin (high/normal/low) -
    - Albumin (high/normal/low) -

    - Hgb (high/normal/low) -
    - Blood glucose (high/normal/low) -
- How does cirrhosis lead to hypoglycemia?

Cholelithiasis & Cholecystitis

- Who is at risk for cholelithiasis?
- How is cholelithiasis different from cholecystitis?
- Describe the pain that patients with cholelithiasis have.
    - What additional manifestations would you expect in a patient with cholelithiasis?
    - What two additional signs would make you think the patient now has cholecystitis? (one is a vital sign; the other is a lab value ☺)

Acute Pancreatitis

- What are the two most common causes of acute pancreatitis?
    -
    -
- Explain the pathophysiology of acute pancreatitis.
- What lab value is elevated and specific to pancreatitis?
- Name three additional signs and symptoms expected in pancreatitis:
    -
    -
    -

ENDOCRINE/SKIN

ADH Disorders

- Complete the chart below with high/low/normal for each of the labs on the left:

	SIADH	DI
ADH		
Na		
Serum Osmolality		
Urine Osmolality		
Urine Output		

- Which disorder will present with hypervolemia? Why?
- Which disorder will present with hypovolemia? Why?

Adrenal Disorders

- What is a common cause of Cushing's Syndrome?
    - Name two specific findings in Cushing's:
        -
        -
    - How does Cushing's Syndrome affect BP and blood glucose levels?
    - How does Cushing's Syndrome affect muscles and bones?
    - What is a major risk of Cushing's Syndrome?
- Who is most at risk for developing Addison's Disease?
    - How does Addison's affect:
        - Blood glucose -
        - BP -
        - Skin -
        - Na and K -
    - What are three major complications of Addison's Disease?
        -
        -
        -

Thyroid Disorders

- Complete the table below:

	Hyperthyroidism	Hypothyroidism
Disease Example		
Temperature Intolerance		
Weight Changes		
Heart Rate		
Peristalsis		
Psychological Symptoms		
Complication		

- Which thyroid disorder(s) have a goiter?
- What labs are involved in thyroid disorders?
    - Why do physicians look at TRH, TSH, and T3/T4?

Diabetes Mellitus

- What is the pathophysiological difference between diabetes mellitus type I and diabetes mellitus type 2?
    - How are the risk factors for development different?
- Explain why patients with type 1 are at an increased risk for DKA.
- What is HHS? What blood glucose levels are typical?
- How do the symptoms of DKA differ from HHS?
    - What ABG results would you expect in a patient with DKA?
        - pH -
        - HCO3 -
        - CO2 -
    - What do patients with DKA have in their urine that patients with HHS don't?

SKIN DISORDERS

Cellulitis

- What is cellulitis?
- List three manifestations you would expect to see in a patient with cellulitis:
    -
    -
    -
- List three patient populations at higher risk for cellulitis development:
    -
    -
    -

Candida Auris

- Candida auris can be a simple _____ infection or become a more serious _____ infection
- Who is most at risk for C. auris infection?

Dermatitis

- What is dermatitis?
- What are the manifestations of dermatitis?
- What is the pathophysiology of…
    - Contact dermatitis:
    - Atopic dermatitis:
- Atopic dermatitis is also known as _____.

Varicella-Zoster

- What two diseases does the varicella-zoster (VZV) cause?

- What is the transmission route for VZV diseases?
- What are the manifestations of…
    - Chickenpox:
    - Shingles:
- List two potential complications of VZV disease:
    -
    -

Melanoma

- List three risk factors for melanoma:
    -
    -
    -
- What acronym is used for mole changes? What do the acronym letters stand for?
    -
    -
    -
    -
    -
- What characteristic of melanoma makes it worse than other skin cancers?

Pressure Ulcers

- Describe the five stages of pressure ulcers/injuries (focus on differences/key words!):
    - Stage 1:
    - Stage 2:
    - Stage 3:
    - Stage 4:
    - Unstageable:
- List five risk factors for pressure ulcer development:
    -
    -
    -
    -
    -
- List three prevention strategies for pressure ulcers (they are nurse-driven!):
    -
    -
    -
- What scale do we use to assess risk for pressure ulcers?

Burns

- Describe the four types of burns (focus on differences/key words!):
    - o
    - o
    - o
    - o
- Name four complications of burns and explain the pathophysiology of each:
    - o
    - o
    - o
    - o
- ***Know your rule of 9s*** ☺